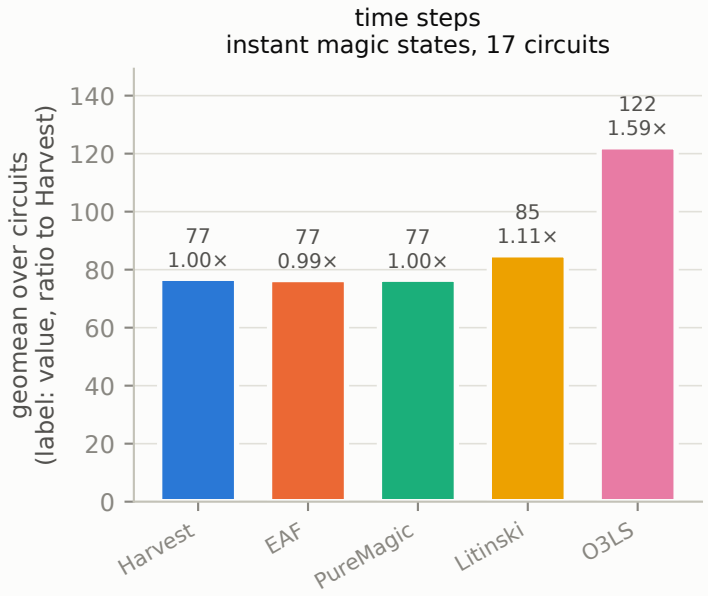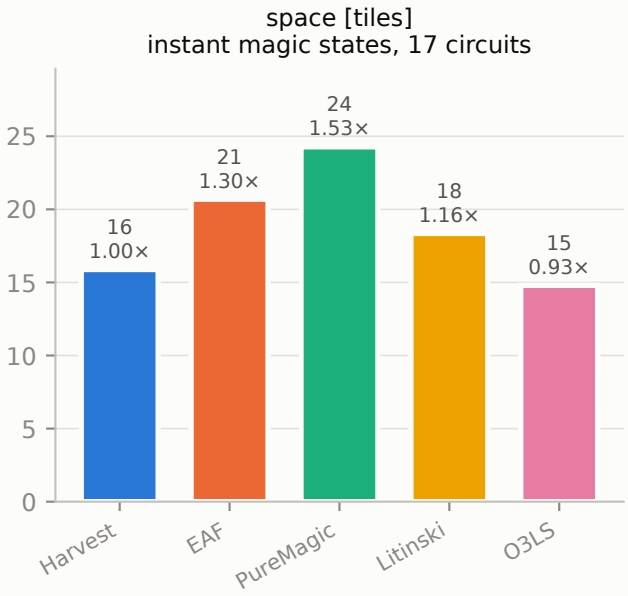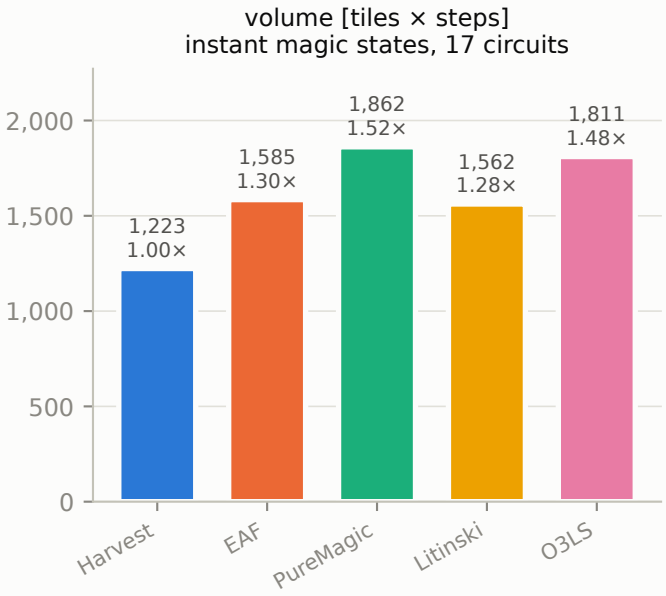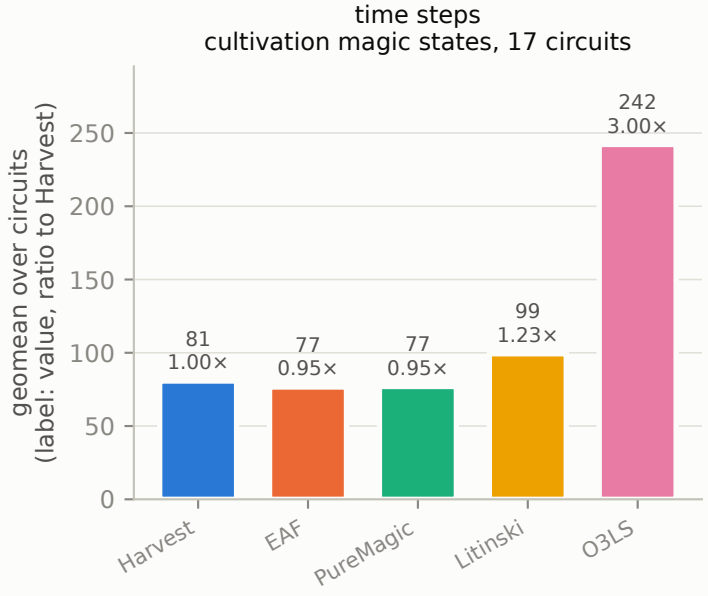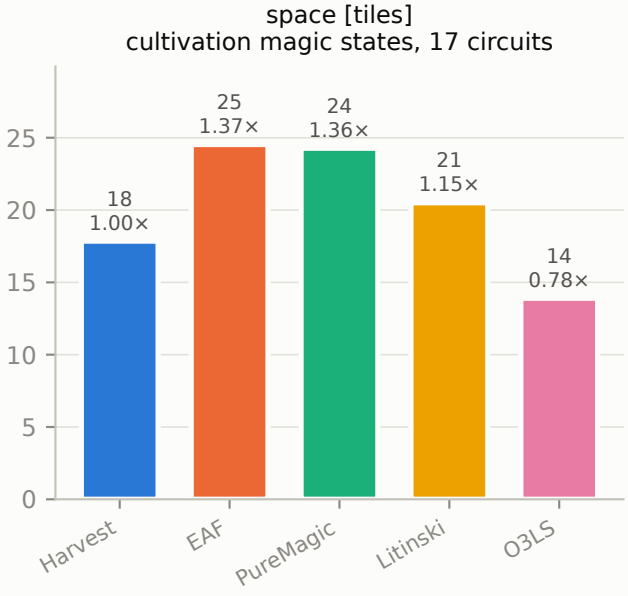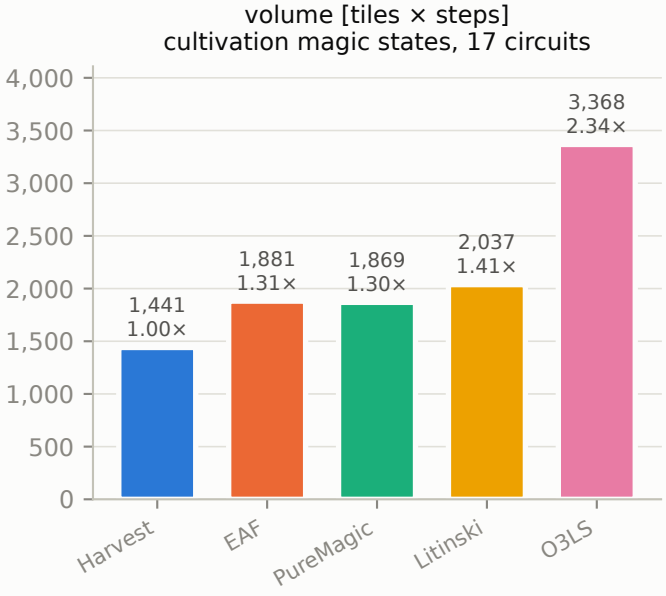other: 17 circuits